

```
imageGen.py  x
7  # WRITE YOUR CODE HERE
8  response = openai.Image.create_variation(
9      image=open("bunny.png", "rb"),
10     n=1,
11     size="1024x1024"
12 )
13 image_url = response['data'][0]['url']
14 #using the url, we use the code below to save it as a file
15 img_data = requests.get(image_url).content
16 with open('image_name_var.png', 'wb') as handler:
17     handler.write(img_data)
```

95% (16:31)

Terminal x

```
codio@cometsugar-horizonlegal:~/workspace$ python imageGen.py
codio@cometsugar-horizonlegal:~/workspace$ ls
bunny.png      edits.py       imageVar.py    original.png    test_image.png
bunny_var.png  exerc.py      masked.png     __pycache__    testVariation.png
edited-copy1.png exercTest.py   mask.png       README.md      varTest.png
edited-copy2.png imageGen.py    mask.py        secret.py      varTests.png
edited.png      image_name_var.png original-copy.png setting.py
codio@cometsugar-horizonlegal:~/workspace$
```

Guide x

Collapse

Text to Image Edits

Dall-E

OpenAI's **DALL-E** text to image is a system that can generate images from textual descriptions, using a deep learning-based approach. The system can generate images that are realistic and editable, and can even create images from scratch, without any training data. Endpoints are often used to retrieve data from a database, create new data, or modify existing data. We have seen that DALL-E models can easily generate images from text. Now we are going to use DALL-E to modify existing data

The Images API provides three methods for interacting with images:

1. Creating images from scratch based on a text prompt
2. Creating edits of an existing image based on a new text prompt
3. Creating variations of an existing image

Generating Image

We are familiar with the steps to generate a new image.

```
#this code generates a new image from a prompt
response = openai.Image.create(
    prompt="digital art of throne room of a magical castle.",
    n=1,
    size="256x256"
)

#from the response generated this puts the url in a separate variable
image_url = response['data'][0]['url']

#using the url, we use the code below to save it as a file
img_data = requests.get(image_url).content
with open('image_name.png', 'wb') as handler:
    handler.write(img_data)
```

TRY IT!

Command was successfully executed.

Click here to refresh your image1

Variation

OpenAI has an image variation endpoint that allows you to generate similar images when given an image as a prompt.

To try it in the last Try It button we created a digital art representation of a castle as `image_name.jpg`. We are going to use that image to generate another similar picture or variation. To avoid rewriting our original file we are going to delete everything up until the comment that says `# Keep CODE ABOVE`. Then we are going to write, the code below to generate a variation.

```
response = openai.Image.create_variation(
    image=open("image_name.png", "rb"),
    n=1,
    size="1024x1024"
)
image_url = response['data'][0]['url']
#using the url, we use the code below to save it as a file
img_data = requests.get(image_url).content
with open('image_name_var.png', 'wb') as handler:
    handler.write(img_data)
```

TRY IT!

Command was successfully executed.

Click here to refresh your image

Image Edits DALL-E

Which of the following methods can be used with the Images API to create a variation of an existing image?

- ☐ `openai.Image.variation`
- ☒ `openai.Image.create_variation`
- ☐ `openai.Image.create_edits`
- ☐ `openai.Image.edit`

The `openai.Image.create_variation()` method is used to create variations of an existing image. It takes an image file as input and generates a new image that is similar to the original one.

Check It!

Next